

Humanvoice

A reader-first revision system for technical writing

Product proposal

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Abstract

Technical authors now have abundant help generating and polishing prose, but they still lack a reliable way to move a complex document from locally fluent to understood and accepted. Humanvoice is a revision system for that gap. It locates diagnosable problems without rewriting automatically, protects equations, numbers, citations, and claims during revision, and ends with a decision from a named human reader. This proposal requests a bounded twelve-week MVP and one-document feasibility run. The build will establish whether the workflow is usable and measurable; it will not be presented as evidence that the product improves expert writing. A later comparative study must earn that claim.

Decision in brief

Build humanvoice as a reader-first revision system for technical documents. The product should help an author find where a draft loses its reader, revise without silently changing its substance, and learn whether the next version actually reduced the work required for acceptance.

The immediate decision is deliberately bounded. Authorize a twelve-week MVP, an evaluation corpus, and one live-document feasibility run. At the end of that work, decide whether to fund a comparative study against the tools and review process already in use. Do not authorize general deployment, and do not claim that humanvoice improves writing before that comparison is complete.

The MVP can be built by one product engineer and one research and evaluation lead, with a part-time domain reviewer. The primary outcome for the eventual study is the number of human-feedback rounds required to reach reader acceptance, provided that reader time and meaning-preservation failures do not increase. If the product cannot improve that outcome, it has not earned a place in the writing process.

The problem worth solving

A technical document can be correct, compile without warnings, and survive several agent reviews while failing in its first few minutes with a human reader. One funding proposal in this workspace did exactly that. Its reader could not state the purpose, did not know why internal file names appeared in the argument, and could not tell what action was being requested. The document

had not failed because of a typo. It had transferred the work of constructing the argument from the author to a scarce expert reader.¹

That transfer has an economic cost. A rejected draft begins another cycle of diagnosis, revision, and rereading. Senior readers spend time rediscovering the same kinds of problems: private project language in public exposition, repetitive paragraph structure, qualifications that never explain a mechanism, unsupported specificity, and missing links between an equation and the decision it is supposed to inform. The cost is not simply author time. Every cycle consumes the attention of the person whose judgment the document was written to support.

Current tools make only parts of this cycle cheaper. Grammar checkers and prose linters find local surface defects. General language models can draft, rewrite, and comment. Citation services can confirm that a DOI exists. Version control can show that text changed. None of these tells us that the intended reader can now reconstruct the argument, and none protects every substantive object while a large passage is recomposed.

Humanvoice is aimed at this missing middle: the journey from a plausible draft to a document a named reader understands and accepts.

Why the obvious alternative stops short

The strongest rival is not another specialist writing product. It is the combination most authors can assemble today: an ordinary prose linter, a general LLM editor, version control, and manual review. Humanvoice should reuse those components where they work and compete on the connection between them.

Revision job	Ordinary toolchain	Humanvoice
Find local style and grammar problems	Strong	Reuses replaceable linters and adds project-register diagnostics
Explain which issue matters most	Inconsistent, often presented as fluent commentary	Ranks located findings and leaves the editorial decision visible
Protect equations, numbers, citations, and claims	Depends on manual comparison	Treats them as protected content and requires an explanation for substantive change
Determine whether the revision helped	Surface scores or an uncalibrated model judgment	Compares reader reconstruction, review burden, and acceptance
Learn across documents	Prompt history and scattered review notes	Retains bounded, privacy-aware revision and reader outcomes

¹The full case record, including the reader’s response and the later repair history, is preserved in [the survey’s internal evidence appendix](#). It is used here as a motivating case, not as external efficacy evidence.

The distinction matters because the evidence for AI assistance is conditional. In a preregistered experiment, generative assistance reduced completion time and improved evaluated output on occupation-specific writing tasks [Noy and Zhang, 2023]. Field evidence also reports time or productivity gains in workplace settings [Dillon et al., 2025, Brynjolfsson et al., 2025]. These studies establish that assistance can change work. They do not estimate the value of this workflow for expert economics writing, and the average effects hide differences by task and worker experience. Assistance can also reduce correctness when a task lies outside the model’s capability frontier [Dell’Acqua et al., 2026].

The appropriate product claim is therefore modest and testable: humanvoice may reduce wasted revision and rereading by connecting diagnosis, protected editing, and reader evidence. The feasibility build is designed to learn whether that mechanism can work. It is not a demonstration that it already does.

The product in use

Humanvoice sits beside the author’s existing editor and repository. It does not replace either. A document moves through four observable steps.

First, the author opens a LaTeX or Markdown draft and chooses the intended reader and decision. Humanvoice parses the document into ordinary prose and protected content. Equations, numbers, labels, citations, quotations, and declared claims retain their source locations.

Second, the system presents a short, ranked set of findings. Some are fully deterministic: a private path appears in the manuscript, five consecutive sections open the same way, a citation has mismatched metadata, or a number changed between versions. Others are editorial questions attached to a span: does this qualification prevent a real misunderstanding, or does it merely protect the author? Is the paragraph missing the mechanism that makes its conclusion follow? Findings remain findings. Humanvoice does not silently edit the source.

Third, the author or an explicitly chosen editor revises the draft. A protected comparison explains what changed and calls attention to any altered equation, number, citation, quotation, or claim. The author may make a substantive change, but it cannot disappear inside a smoother sentence.

Finally, a low-context reviewer receives the rendered document without the repository history that made its private language seem familiar. The reviewer answers questions tied to the document’s purpose: What problem is being solved? What is proposed? Why is the obvious alternative insufficient? What evidence matters? What decision is requested? A named human remains the final authority. A missing answer sends the document back for substantive revision, not another round of synonym replacement.

Revision flow

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source -> format-aware parse -> located findings -> author revises
      |                                     |
      +--> protected content -----+
                                   |
                                   protected comparison
                                   |
                                   low-context read -> human decision

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This design keeps two records for good reason. The reader sees only the document. The project retains the minimum provenance needed to reproduce a diagnostic, understand a revision, and evaluate the workflow. Internal IDs, search administration, and review machinery never become manuscript prose.

What the evidence changes

The literature does not point toward an autonomous rewriting system. It points toward a bounded editor with direct evaluation of the resulting revision.

Feedback must be judged by the next draft

Writing feedback can sound specific while missing the most important problem. On a test set of 1,300 deliberately corrupted stories, models often produced accurate local comments but failed to identify the central defect or choose appropriately between criticism and praise [Rashkin et al., 2025]. Work on an economic essay task similarly evaluates feedback through the revisions it induces rather than through the fluency of the comment itself [Nair et al., 2024]. Humanvoice therefore links each finding to a source span, records whether the writer acted on it, and evaluates what changed in the next version.

Two recent systematic reviews reinforce the boundary. Across their respective evidence bases, benefits are more consistent for grammar, process, coherence, and surface accuracy than for argumentation, discourse organization, or creativity [Meng et al., 2026, Abudalfa and Barrot, 2026]. Humanvoice can automate some measurements at the surface. It must measure higher-order understanding with the intended readers.

Model judgments need calibration

Model judges are useful screening instruments, but their preferences change with answer order, length, and model family [Zheng et al., 2023, Panickssery et al., 2024, Dubois et al., 2024]. They also tend to be more generous than expert writers when judging creative prose [Chakrabarty et al., 2024]. Any model-assisted review in humanvoice will therefore address one construct at a time, randomize pair order, control for length, use a different model family from the editor, and report calibration against human labels. It may direct attention. It may not accept a document.

Assistance can change trust and voice

Disclosure of AI assistance can affect reader ratings and increase disagreement among readers [Li et al., 2024]. Assistance can also increase individual performance while reducing the diversity of a collection [Doshi and Hauser, 2024, Padmakumar and He, 2024]. These are product outcomes, not matters of cosmetic style. Humanvoice will retain the disclosure condition used in evaluation and monitor convergence across a sufficiently large corpus. It will never infer authorship from an individual document or optimize text to evade a detector.

Citation identity is not claim support

Generated scholarly references can be nonexistent or mismatched [Mugaanyi et al., 2024]. A metadata service can establish that a paper and DOI match. Only inspection of the paper can

establish that it supports the sentence in which it is cited. Humanvoice separates those jobs: automatic identity resolution followed by a human source-to-claim check for important claims.

These findings narrow the design and the promise. They justify local measurement, protected revision, explicit uncertainty, and direct reader outcomes. They do not justify a single quality score, an automatic rewrite, an authorship detector, or an efficacy claim before the pilot.

What we will build

The MVP has four product capabilities and one evaluation asset.

The document core reads LaTeX and Markdown while preserving source locations and protected regions. Pandoc and pylatexenc are the first parser candidates because they expose structured documents through mature, documented interfaces [Pandoc maintainers, 2026, pylatexenc maintainers, 2026]. They will be tested on unknown macros, nested environments, comments, mathematics, citations, malformed input, and round-trip preservation. A parser is selected only after those fixtures pass.

The diagnostic layer locates private project register, structural repetition, defensive prose candidates, citation identity problems, and build or rendering defects. Vale and existing local rules are plausible hosts for deterministic prose checks [Vale maintainers, 2026]. Every result carries its source span and a clear question or measurement. No diagnostic command mutates its input.

The protected comparison reports changed equations, numbers, labels, citations, quotations, and declared claims. latexdiff may provide a useful rendered comparison, but the invariant check remains structural [latexdiff maintainers, 2026]. An unexplained protected change stops that revision from being presented to a reader.

The review workspace shows a clean rendered draft, asks a short document-specific rubric, and records the minimum decision evidence. It does not expose the author's plans or audit records. Model-generated suggestions are optional, identified, and reversible.

The evaluation corpus begins with existing rejected and repaired documents for which reader verdicts or revision histories are already available. It will add adjudicated labels for defect location, issue importance, protected changes, and acceptance. Tuning and held-out documents will be separated before package or model results are reported.

The detailed command contracts and sixteen implementation requirements remain available in the requirements trace. They are engineering inputs, not the public explanation of the product.

How we will know whether it works

The evaluation has two stages because feasibility and efficacy answer different questions.

The first live document tests whether the pieces work together. Can the parser protect the document? Do findings point to the right source spans? Can an author understand and dismiss a false positive? Does the comparison expose meaning changes? Can a reader complete the rubric without seeing private project machinery? Does the workflow respect privacy and disclosure choices? One document can reveal integration failures and unacceptable burden. It cannot estimate an effect.

If feasibility passes, the project will preregister a paired or stepped comparison against the current workflow: existing linters, ordinary LLM editing where the author already uses it, version control, and manual review. Documents will be selected before outcomes are known. Reader blinding will be used where practical. The sample size will be chosen by simulating power or precision from the observed distribution of baseline revision rounds and reader clustering, not by choosing a convenient number of documents.

The primary estimand is

$$\Delta r = E[r_{\text{humanvoice}} - r_{\text{current}}],$$

where r is the number of human-feedback rounds to acceptance. Deployment requires evidence that $\Delta r < 0$ without an increase in meaning- preservation failures or total reader time. Reader reconstruction accuracy, substantive revision rate, false-positive burden, editing time, suggestion acceptance, and cross-document convergence are explanatory outcomes. None may replace the primary endpoint.

Results will be reported separately by writer experience, document genre, and reader role. This is not optional subgroup decoration. Field evidence shows that average assistance effects can conceal smaller gains or harm among the most experienced workers [Brynjolfsson et al., 2025].

The project stops or changes direction when any of the following occurs:

- unexplained changes to protected content exceed the prespecified tolerance;
- diagnostic false positives consume more review time than they save;
- privacy or disclosure requirements cannot be met;
- a feasibility user cannot understand or control the workflow; or
- reader acceptance does not improve, or becomes slower, after two comparative pilot cycles.

Delivery and investment

The twelve-week program ends with a working feasibility system, not a claim of market readiness.

	Weeks	Work	Reader-visible result
	1–2	Freeze the baseline corpus, reader rubric, protected-content model, and parser fixtures	One representative document parses and renders with protected spans intact
	2–5	Implement located diagnostics, citation identity checks, and protected comparison	An author can inspect findings and revise without hidden source mutation
	3–7	Label tuning and held-out documents; benchmark parser and diagnostic candidates	Package decisions are supported by observed behavior rather than popularity

	Weeks	Work	Reader-visible result
	6–9	Integrate the author and reader workflow with local-first data handling	One document can move from source through revision to a recorded reader response
	9–12	Run the live feasibility document, measure burden and failures, and price the next study	Proceed, revise, or stop recommendation with a preregisterable efficacy design

The planning assumption is one product engineer, one research and evaluation lead, and a part-time economics or scholarly-writing reviewer. Twelve weeks covers the MVP and feasibility case. The efficacy study’s duration depends on the required precision and the arrival rate of suitable live documents.

The investment case should be calculated from project data. Let r_0 and r_1 be revision rounds under the current and proposed workflows, t_0 and t_1 the expert-reader hours per round, c_r the loaded cost of that time, N the annual number of relevant documents, and C the annualized build and maintenance cost. The measurable reader-time component of annual value is

$$V = N(r_0t_0 - r_1t_1)c_r - C.$$

Quality, author time, and avoided meaning failures may add value, but they should enter the decision only when the pilot measures them. No published study supplies the project-specific inputs to this equation.

Risks and limits

The most serious product risk is meaning drift. Smoother prose is a loss if it changes a model, number, source, or conclusion. Protected comparison and human source checks are therefore part of the core, not later compliance features.

A second risk is alert fatigue. A diagnostic that is technically correct but rarely useful can make the workflow slower and teach authors to ignore it. The held-out benchmark will report precision by category and the time required to review findings. Low-value rules will be removed rather than defended by their coverage.

A third risk is convergence on a house style. Humanvoice is not meant to make every economist sound alike. It reports patterns and asks questions; it does not automatically normalize wording. Aggregate monitoring will look for loss of lexical and structural diversity, while individual documents will never receive authorship labels.

Privacy and disclosure create a fourth boundary. Protected manuscripts remain local by default. Any remote model use must identify the provider, purpose, retention terms, and approved content. The reader study must use the disclosure condition intended for actual use because disclosure itself can affect reader response.

Finally, the supporting evidence is not yet a completed systematic review. The current audit has broad public discovery and verified bibliographic identities, but independent screening, specialist database coverage, full-text verification, critical appraisal, held-out package benchmarks, and external review remain incomplete. The detailed status is reported in the [evidence audit](#). This limitation blocks strong literature and efficacy claims. It does not prevent a reversible build whose purpose is to measure feasibility.

Decision requested

Authorize the twelve-week MVP and one live-document feasibility run under the resource assumptions and stop rules above. The authorized deliverables are:

1. a local-first revision system for LaTeX and Markdown;
2. located, non-mutating diagnostics and protected comparison;
3. a small adjudicated corpus with frozen tuning and held-out partitions;
4. one end-to-end feasibility result, including burden and failure evidence;
5. a costed, power-justified design for the comparative study; and
6. a recommendation to proceed, revise, or stop.

This decision does not authorize autonomous rewriting, authorship detection, general deployment, or a claim that humanvoice improves expert writing. It authorizes the smallest build that can determine whether those further investments are warranted.

Technical appendix

Product boundary

The MVP accepts local LaTeX and Markdown sources and emits findings, protected comparisons, rendered review material, and bounded evaluation events. It does not generate claims or citations, classify authorship, evade detectors, upload manuscripts without approval, or accept a document on behalf of its reader.

The stable CLI concepts from the founding survey remain useful implementation interfaces:

Command	Contract
<code>hv leak</code>	Locate private project register and disclosure problems without editing
<code>hv rhythm</code>	Report sentence, paragraph, and section-shape measurements against genre configuration
<code>hv defensive</code>	Return contextualized candidate passages and editorial questions
<code>hv diff</code>	Explain changes to protected content between document versions
<code>hv cite</code>	Resolve bibliographic identity and prepare a separate source-to-claim review queue

Command	Contract
<code>hv gate</code>	Build and render the document, report reference and layout failures, and expose pages for inspection
<code>hv drift</code>	Measure population-level convergence on sufficiently large corpora; refuse individual attribution

The names are less important than the shared contracts: immutable input, stable source locations, structured output, no silent source mutation, and no diagnostic promotion of a document.

Core records

Implementation needs six small records: a document version with protected spans; a located diagnostic finding; an explained change; a citation identity and claim anchor; a reader response tied to one rendered version; and a privacy-bounded interaction event. Their complete fields and evidence links are maintained in the machine-readable requirements rather than repeated in the main proposal.

Supporting scholarship and audit

The full literature and software treatment is in [the supporting survey](#). Search histories, screening queues, evidence extraction, package inventory, manifests, and unresolved review work remain under [docs/survey/audit/latest/](#). Those records protect the product’s claims and implementation choices. They are not the product proposal’s narrative.

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